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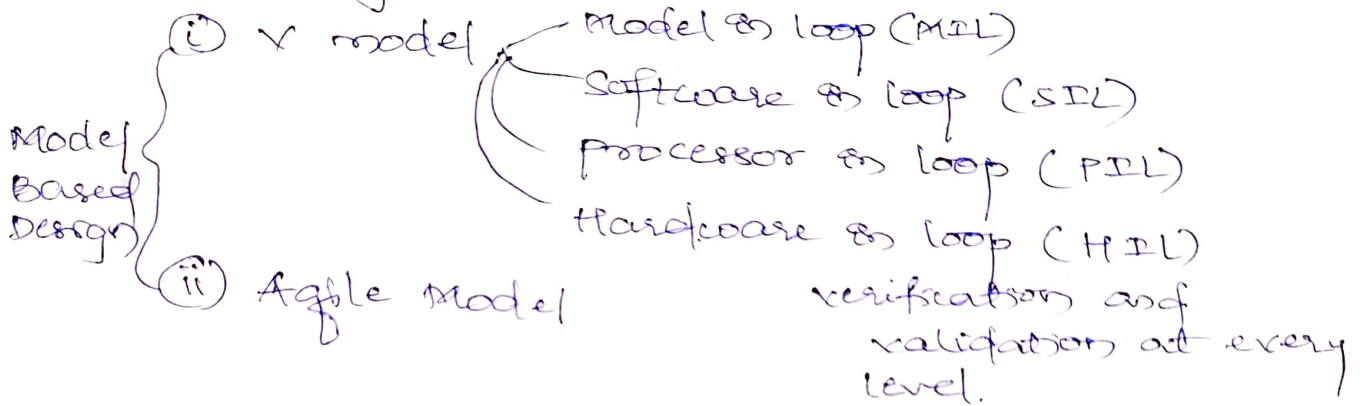
Chapter 1 Automotive Electronics
(25EECC305)

Syllabus

- * Overview of Automotive Industry
- * Vehicle Functional Domains
- * Automotive Mechanical systems
- * ECU Design Cycle
 - Model Based Design
 - V-model
 - Agile
- * Fundamentals of Electric Vehicles
 - (i) Drive cycles
 - (ii) EV Drive Train
 - (iii) EV Batteries
 - (iv) Types of EV

Notes includes

- ① Automotive supply chain
- ② Automotive value chain
- ③ functional domains
- ④ Automotive Mechanical systems
- ⑤ ECU Design



- ⑥ Electric vehicles

Automotive Fundamentals and Electronics

Automotive Electronics

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Automobile Electronics

used

to replace mechanical support
composed of all computer controlled, sensor based
and electrical systems in modern vehicles.

Why need?

① Engine Control Module (ECM) → Control the temperature inside engine.

fuel efficiency → Exhaust particle → High power → higher Torque → Control knocking

② Vehicle
slip control

over steering control → under steering control
ABS, TCS and ESP
vehicle Roll over control → Reduce stopping distance → Brake distribution

1. ECM

ECU → BCM

2. ABS, TCS & ESP

ABS → Anti-lock Braking system

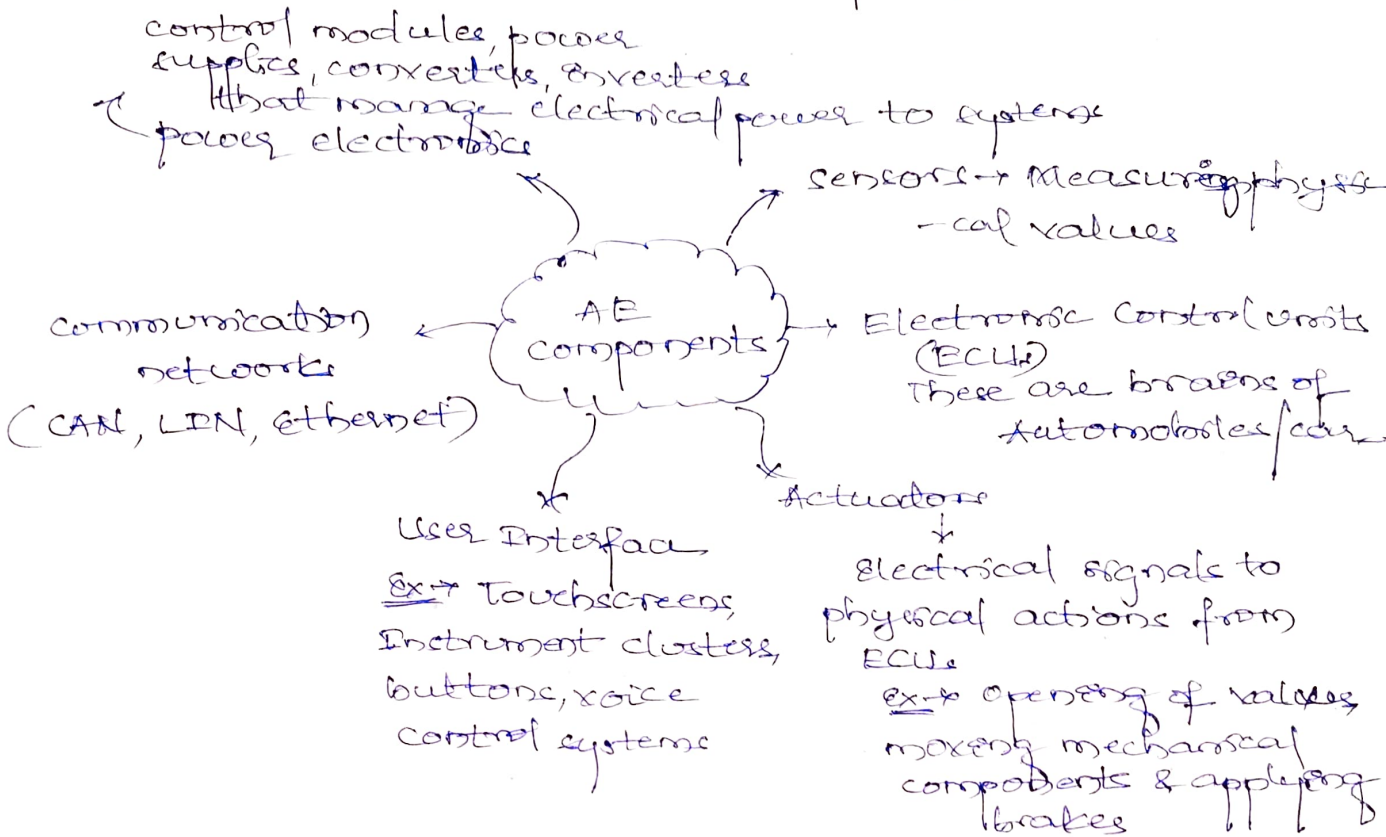
TCS → Traction control system

ESP - Electronic stability program

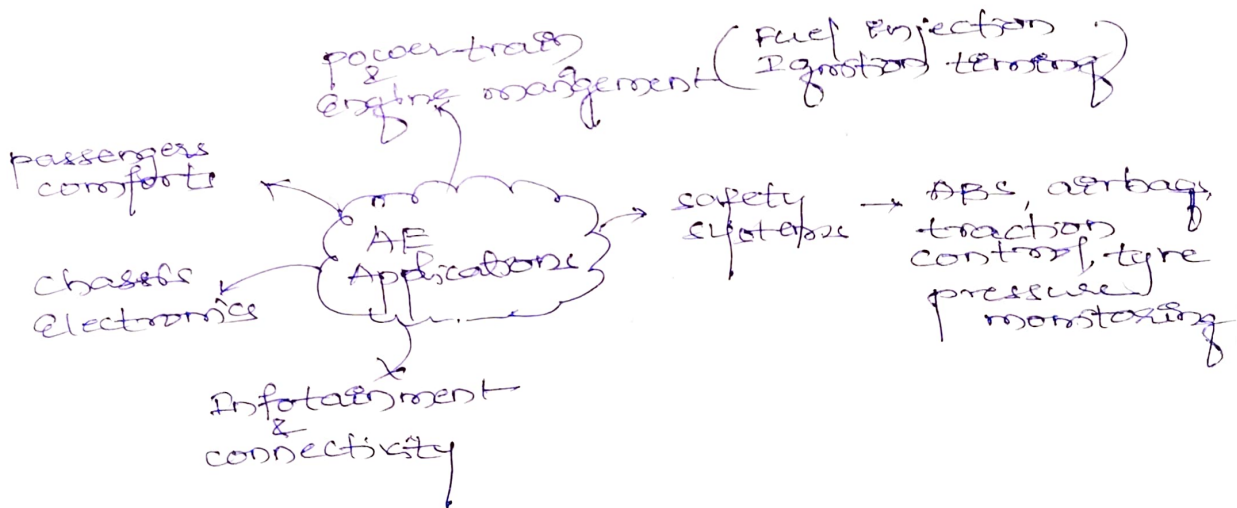
③ Body Control Module

Vehicle Interior Monitoring → BCM → Automatic door lock
Vehicle Exterior Monitoring → Remote key → power windows

Automotive Electronics Components



Applications of Automotive Electronics



Overview of ABS, TCS, ESP and ADAS

ABS (Anti-lock Braking System)
Braking / Deceleration (purpose) → prevents wheels from locking during hard braking to maintain steering

TCS (Traction control system)
→ prevents driven wheels from spinning excessively during acceleration.

ESP (Electronic stability program)
purpose → counteracts skidding, understeering & oversteering to keep car on its intended path.

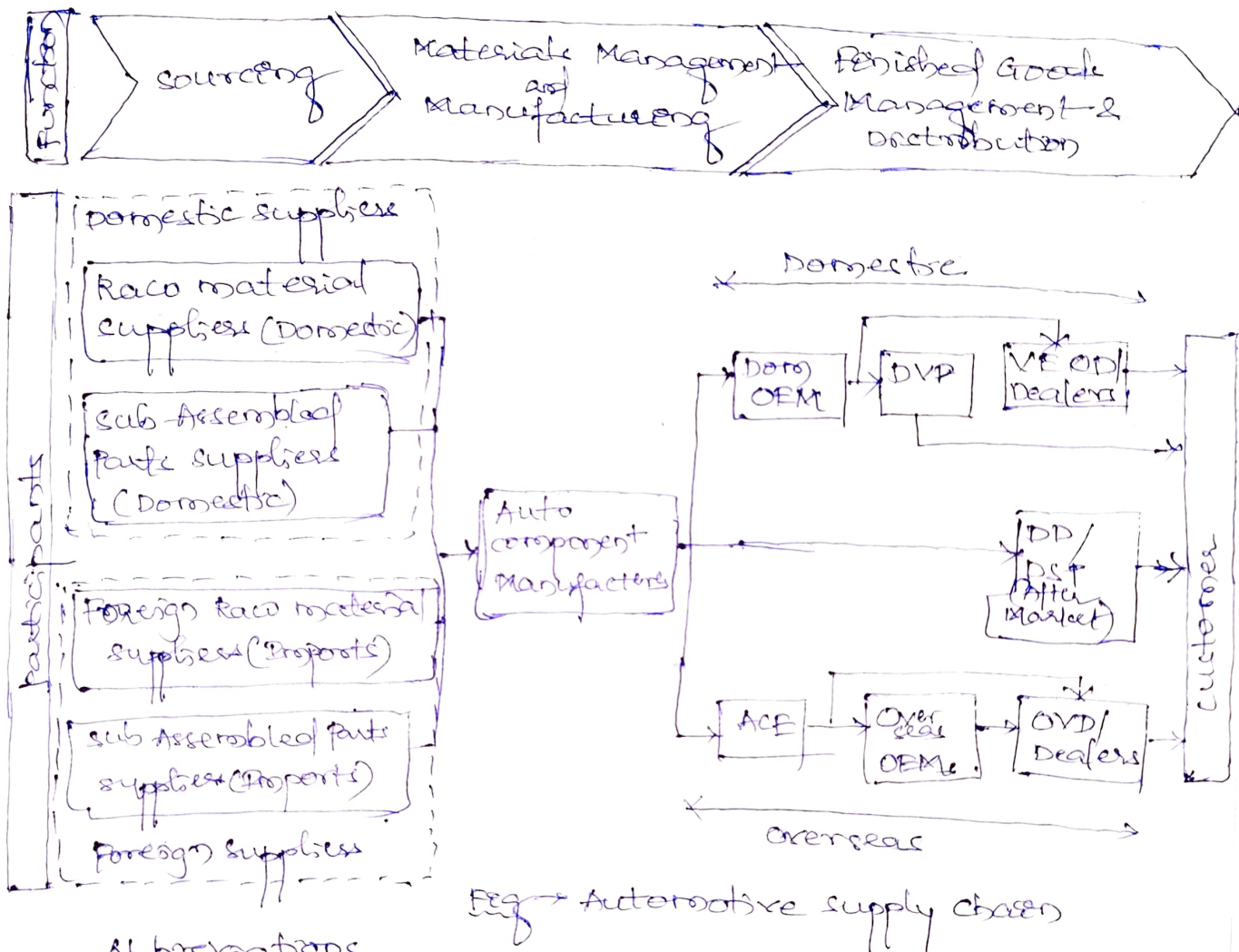
ADAS (Advanced Driving Assistance System)
purpose → Safety and convenience
→ Adaptive cruise control
→ Lane departure warnings
→ Blindspot detection
→ Automatic Emergency braking

Note → You will study more about ADAS in Unit-3)

Automotive supply chain

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The automotive supply chain is a vast, intricate, global network of manufacturers, suppliers and service providers that work together to bring vehicles to the market.



Abbreviations

OEM → Original Equipment Manufacturers

DVD → Domestic Vehicle Dealers

VEOD → Vehicle Exports to overseas distributors

DD/DCP → Domestic Distributors/Dealers of spare parts

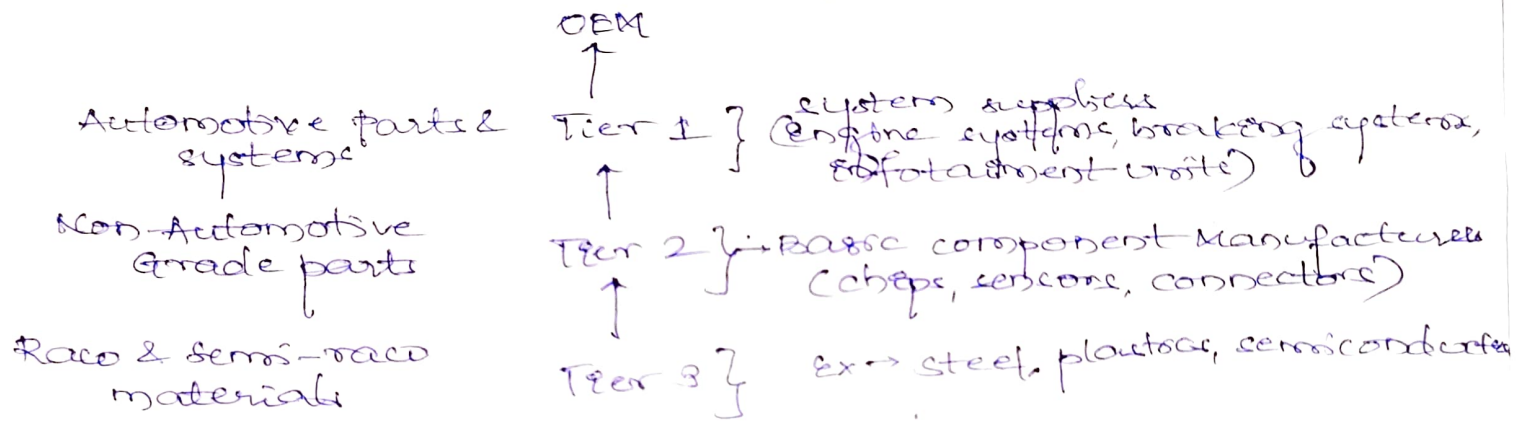
ACE → Auto Component exports

OVD → Overseas Vehicle Distributors

④ Overseas OEMs and Dealers → Car parts and

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- * Automotive supply chain consists of multiple interconnected stages involved in designing, manufacturing, assembling and delivering vehicles and components.
- * Includes raw material suppliers, component manufacturers, system integrators, vehicle assemblers, distributors and dealerships.



- * This refers to the process of how vehicle is made, starting from raw materials to final product till reaching the customer.
 - * Has multiple participants involved in sourcing, manufacturing and distribution.
- process is divided into three main parts or stages
- ① sourcing (Getting materials)
 - ② Material Management and manufacturing
→ making vehicle parts and assembling them
 - ③ Finished Goods Management and Distribution
→ delivering vehicles to the customers.

(i) Sourcing

* Basically getting materials i.e., before manufacturing the vehicle companies need raw materials and parts and these come from different sources.

(a) Domestic suppliers / local suppliers

* Provide a raw material like steel, plastic, rubber and glass. Also supply sub-assembly parts.

(b) Foreign suppliers

* Import from other countries
* Sometimes manufacturers import raw materials if they are cheaper or in better quality.
* They import pre-assembled parts, that are not available in local.

(ii) Material Management and manufacturing

* Once all raw materials and parts are collected, auto component manufacturers create different car part such as engine, transmissions and braking systems, those components are then sent to

(a) Domestic OEMs → companies assemble vehicle using the parts.

(b) Auto component exports → They export some parts to other countries

(iii) Finished Goods Management & Distribution

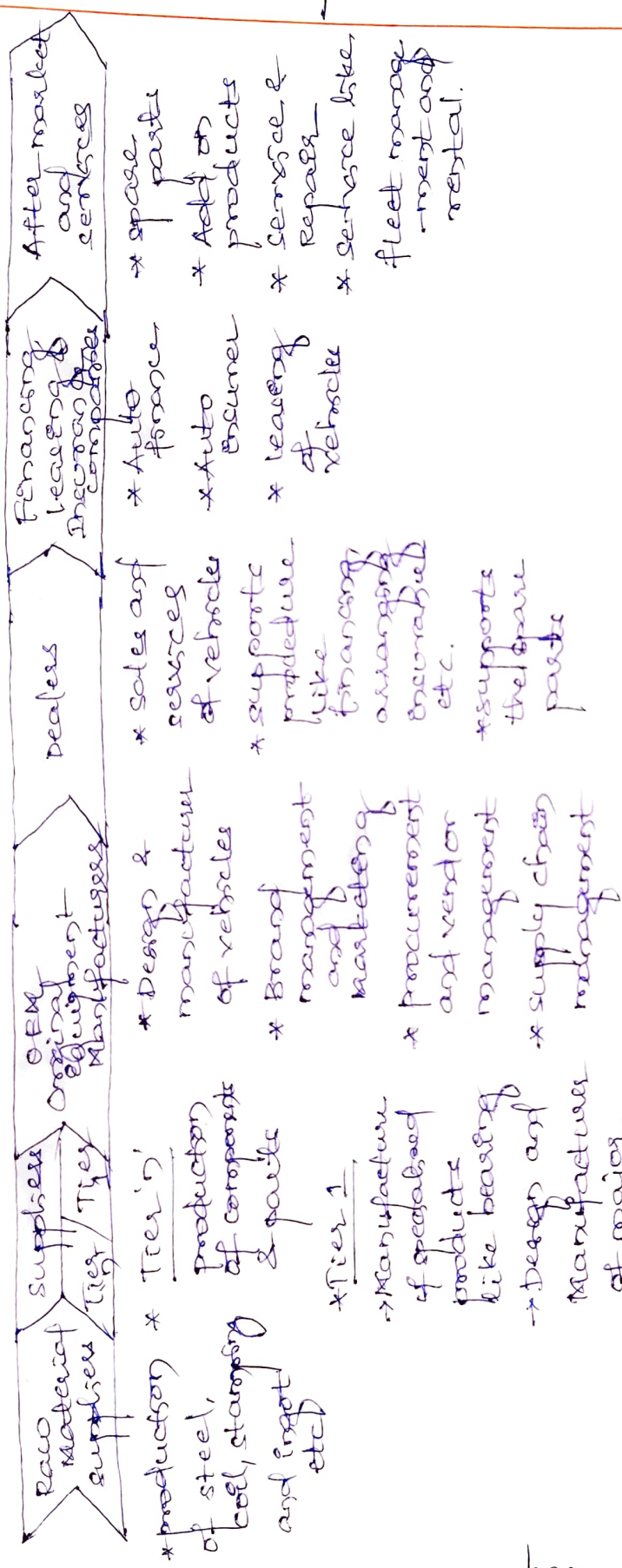
After manufacturing, vehicles are sent to the customers through different channels

(a) Domestic vehicle dealers → car sent to local car showrooms for sales.

(b) Vehicle Export to Dealers → some vehicles are shifted to other countries for sales.

(c) Domestic distribution → spare parts are sent for repair and replacement purpose.

Automotive Value chain



* Automotive value chain :- process of making and selling vehicles

Automotive supply chain has sub-divisions

- (i) Raw material suppliers
- (ii) Suppliers
- (iii) OEM
- (iv) Dealers
- (v) Financing, leasing and insurance companies
- (vi) Aftermarket and services.

(7) Overseas OEMs and Dealers -> Car parts and vehicles are sold to foreign distributors and a dealerships.

i) Raw Material Suppliers

- * These are the companies that provide the basic materials needed to make cars, materials like steel, Aluminium and other materials.

ii) Suppliers (Tier "n" and Tier "1")

- * Tier "n" → make small parts of components
- * Tier "1" → now, tier "n" will produce the small parts components, these are assembled into bigger systems like car interiors, seating, and even engines.

iii) OEM (Original Equipment Manufacturers)

- * These are the car companies that take all the parts and put them together to make a final vehicle
- * They handle branding, marketing and supply chain management

iv) Dealers

- * These are the businesses that sell the cars to customers.
- * They offer financing, insurance and accessories.

v) Financing, Leasing and Insurance Companies

- * Obviously the margin and rate/cost of the cars are maximum, so the customers preferring lease to buy cars.
- * These companies help people buy or lease cars by offering loans and insurance.

vi) Aftermarket & services

- * Once the car is sold, these businesses provide spare parts, repairs and rental services.